

Лабораторная работа №14

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Цель работы

Приобретение практических навыков администрирования сетевых подсистем

На сервере установим необходимые пакеты samba, samba-client и cifs-utils (рис. @fig-1):

```

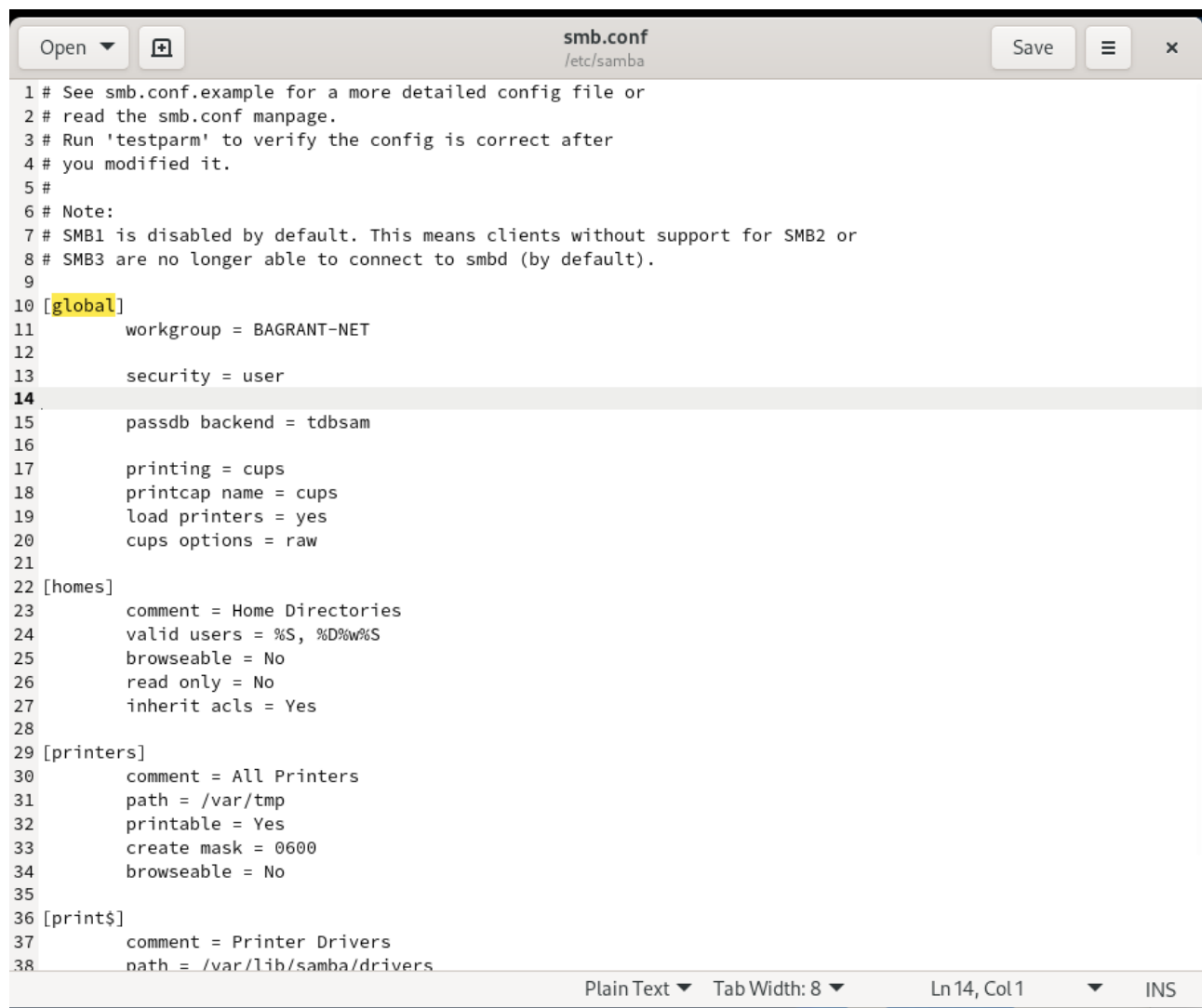
root@server:~
agrant@server ~]$ dnf -y install samba samba-client cifs-utils
Error: This command has to be run with superuser privileges (under the root user on most systems).
agrant@server ~]$ sudo -i
root@server ~)# dnf -y install samba samba-client cifs-utils
Rocky Linux 9 - BaseOS                    5.6 kB/s | 4.3 kB    00:00
Rocky Linux 9 - AppStream                  5.7 kB/s | 4.8 kB    00:00
Rocky Linux 9 - Extras                    7.5 kB/s | 3.1 kB    00:00
Dependencies resolved.
=====
Package                               Arch      Version              Repository           Size
=====
Installing:
cifs-utils                           x86_64    7.2-1.el9            baseos               113 k
samba                                x86_64    4.22.4-12.el9_7      baseos               930 k
samba-client                         x86_64    4.22.4-12.el9_7      appstream            760 k
Installing dependencies:
libnetapi                            x86_64    4.22.4-12.el9_7      baseos               142 k
samba-common-tools                   x86_64    4.22.4-12.el9_7      baseos               483 k
samba-dcerpc                         x86_64    4.22.4-12.el9_7      baseos               717 k
samba-ldb-ldap-modules               x86_64    4.22.4-12.el9_7      baseos                34 k
samba-libs                           x86_64    4.22.4-12.el9_7      baseos               123 k
Transaction Summary
=====
Install 8 Packages

Total download size: 3.2 M
Installed size: 12 M

```

Рис. 1: Этап 1

Создадим группу `sambagroup` для пользователей, добавим пользователя и создадим общий каталог (рис. @fig-2):



```
1 # See smb.conf.example for a more detailed config file or
2 # read the smb.conf manpage.
3 # Run 'testparm' to verify the config is correct after
4 # you modified it.
5 #
6 # Note:
7 # SMB1 is disabled by default. This means clients without support for SMB2 or
8 # SMB3 are no longer able to connect to smbd (by default).
9 #
10 [global]
11     workgroup = BAGRANT-NET
12
13     security = user
14
15     passdb backend = tdbsam
16
17     printing = cups
18     printcap name = cups
19     load printers = yes
20     cups options = raw
21
22 [homes]
23     comment = Home Directories
24     valid users = %S, %D%w%S
25     browseable = No
26     read only = No
27     inherit acls = Yes
28
29 [printers]
30     comment = All Printers
31     path = /var/tmp
32     printable = Yes
33     create mask = 0600
34     browseable = No
35
36 [print$]
37     comment = Printer Drivers
38     path = /var/lib/samba/drivers
```

Рис. 2: Этап 2

В файле конфигурации `/etc/samba/smb.conf` изменим параметр рабочей группы (рис. @fig-3):

В конце файла добавим раздел с описанием общего доступа к разделяемому ресурсу (рис. @fig-4):

Убедимся, что мы не сделали синтаксических ошибок в файле `smb.conf`, используя команду `testparm` (рис. @fig-5):

Запустим демон Samba и посмотрим его статус (рис. @fig-6):

```
Complete!
[root@server ~]# groupadd -g 1010 sambagroup
[root@server ~]# usermod -aG sambagroup user
[root@server ~]# mkdir -p /srv/smbashare
[root@server ~]# gedit /etc/samba/smb.conf

(gedit:9023): dconf-WARNING **: 15:20:19.112: failed to commit changes to dconf: Failed
to execute child process "dbus-launch" (No such file or directory)

(gedit:9023): dconf-WARNING **: 15:20:19.121: failed to commit changes to dconf: Failed
to execute child process "dbus-launch" (No such file or directory)

(gedit:9023): dconf-WARNING **: 15:20:19.620: failed to commit changes to dconf: Failed
to execute child process "dbus-launch" (No such file or directory)

(gedit:9023): dconf-WARNING **: 15:20:19.620: failed to commit changes to dconf: Failed
to execute child process "dbus-launch" (No such file or directory)
```

Рис. 3: Этап 3

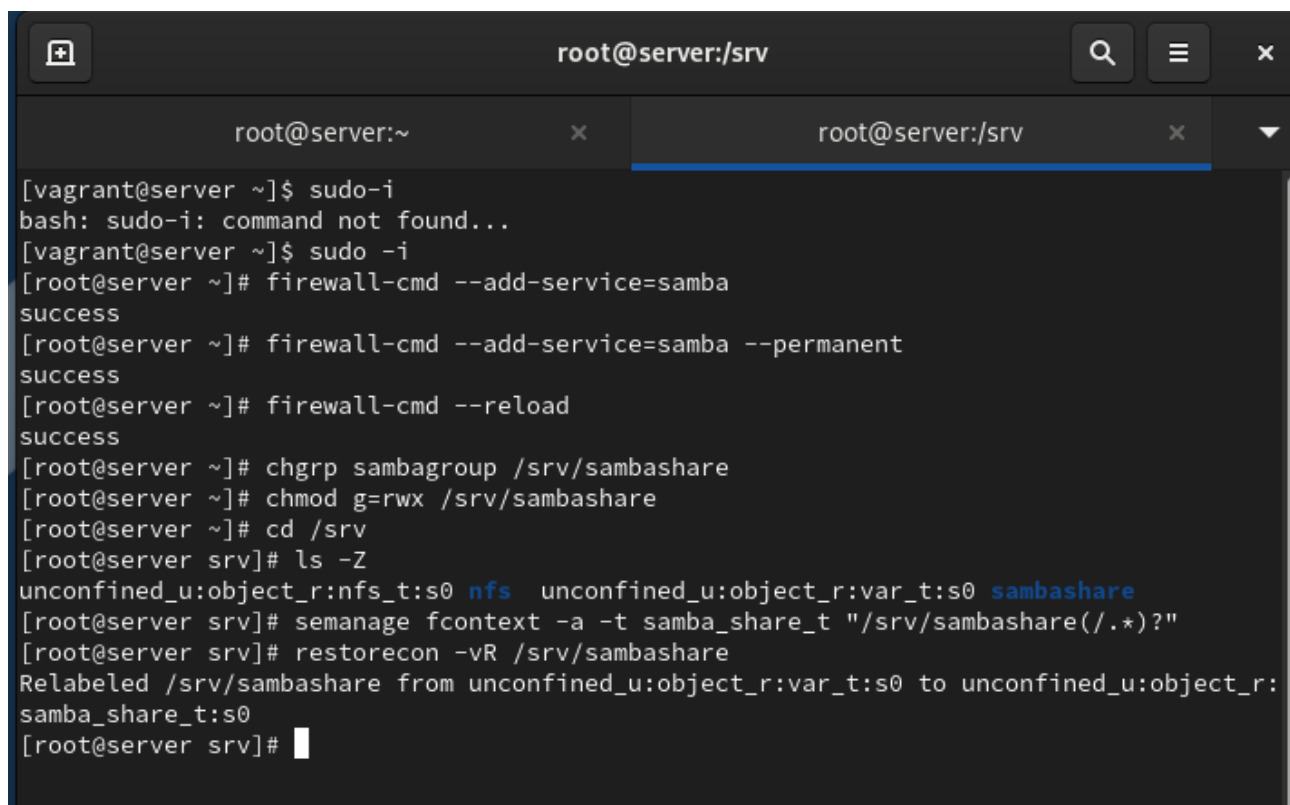
```
35 [print$]
36     comment = Printer Drivers
37     path = /var/lib/samba/drivers
38     # printadmin is a local group
39     write list = printadmin root
40     force group = printadmin
41     create mask = 0664
42     directory mask = 0775
43 [smbashare]
44     comment = My Samba Share
45     path = /srv/smbashare
46     write list = @sambagroup
```

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Рис. 4: Этап 4

```
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>Samba</short>
  <description>This option allows you to access and participate in Windows file and printer sharing networks. You need the samba package installed for this option to be useful.</description>
  <include service="samba-client"/>
  <port protocol="tcp" port="139"/>
  <port protocol="tcp" port="445"/>
</service>
/usr/lib/firewalld/services/samba.xml (END)
```

Рис. 5: Этап 5



```
root@server:/srv
root@server:~
[vagrant@server ~]$ sudo-i
bash: sudo-i: command not found...
[vagrant@server ~]$ sudo -i
[root@server ~]# firewall-cmd --add-service=samba
success
[root@server ~]# firewall-cmd --add-service=samba --permanent
success
[root@server ~]# firewall-cmd --reload
success
[root@server ~]# chgrp sambagroup /srv/sambashare
[root@server ~]# chmod g=rwx /srv/sambashare
[root@server ~]# cd /srv
[root@server srv]# ls -Z
unconfined_u:object_r:nfs_t:s0 nfs unconfined_u:object_r:var_t:s0 sambashare
[root@server srv]# semanage fcontext -a -t samba_share_t "/srv/sambashare(/.*)?"
[root@server srv]# restorecon -vR /srv/sambashare
Relabeled /srv/sambashare from unconfined_u:object_r:var_t:s0 to unconfined_u:object_r:
samba_share_t:s0
[root@server srv]#
```

Рис. 6: Этап 6

Для проверки наличия общего доступа попробуем подключиться к серверу с помощью smbclient (рис. @fig-7):

Посмотрим файл конфигурации межсетевого экрана для Samba и настроим его (рис. @fig-8):

Настроим права доступа для каталога с разделяемым ресурсом, контекст безопасности SELinux и добавим пользователя в базу Samba (рис. @fig-9):

На клиенте установим необходимые пакеты samba-client и cifs-utils (рис. @fig-10):

На клиенте настроим межсетевой экран, создадим группу и изменим рабочую группу (рис. @fig-11):

Создадим файл smbusers в каталоге /etc/samba/ для настройки работы с Samba с помощью файла учётных данных (рис. @fig-12):

Добавим строку в /etc/fstab и подмонтируем общий ресурс (рис. @fig-13):

```
root@client:~  
[vagrant@client ~]$ sudo -i  
[root@client ~]# dnf -y install samba-client cifs-utils  
Rocky Linux 9 - BaseOS      8.6 kB/s | 4.3 kB    00:00  
Rocky Linux 9 - BaseOS      6.5 MB/s | 16 MB     00:02  
Rocky Linux 9 - AppStream    15 kB/s | 4.8 kB     00:00  
Rocky Linux 9 - AppStream    4.4 MB/s | 17 MB     00:03  
Rocky Linux 9 - Extras       7.3 kB/s | 3.1 kB     00:00  
Rocky Linux 9 - Extras      33 kB/s | 17 kB     00:00  
Dependencies resolved.  
=====
```

Package	Arch	Version	Repository	Size
---------	------	---------	------------	------

```
=====
```

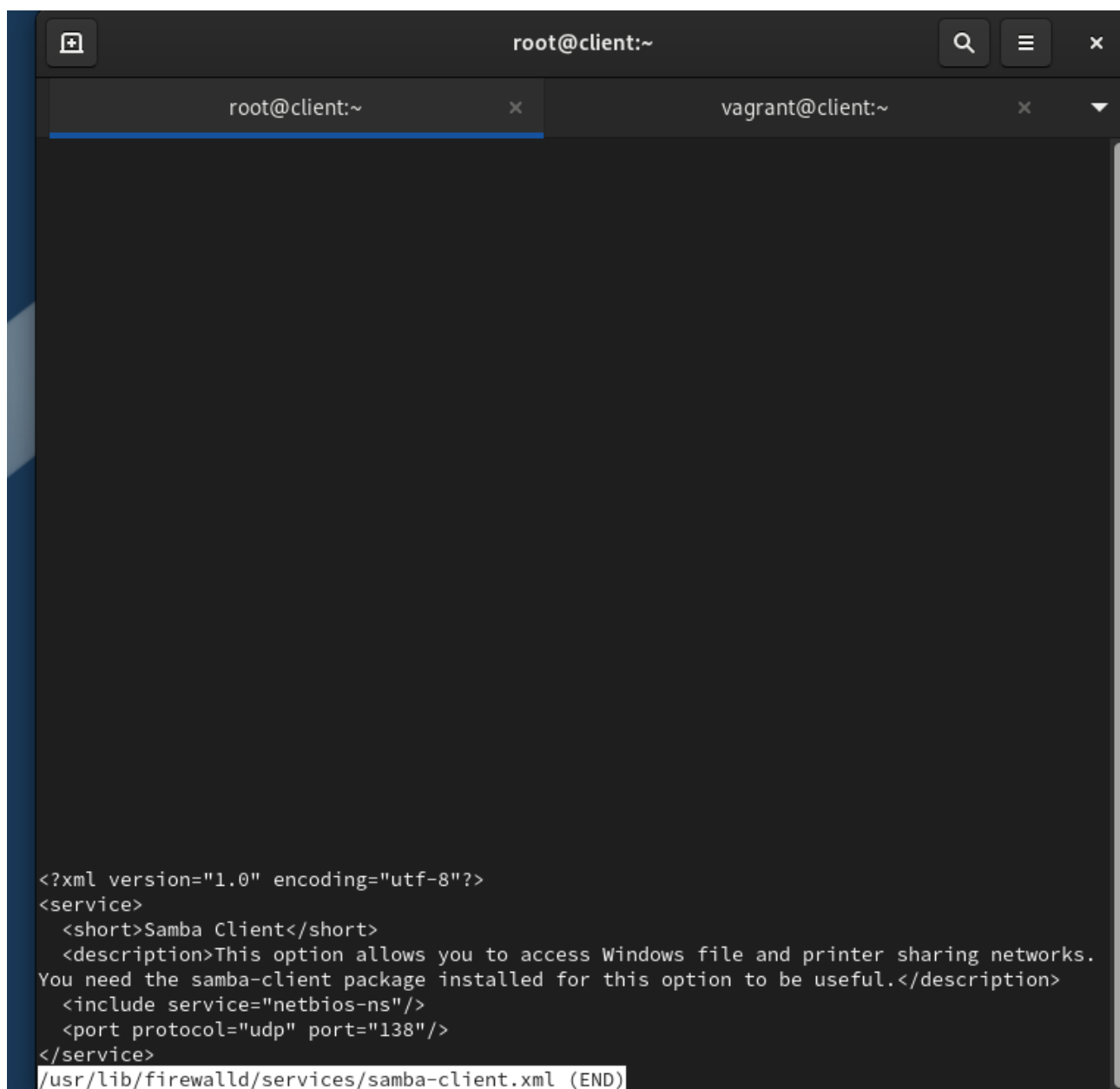
Installing:

cifs-utils	x86_64	7.2-1.el9	baseos	113 k
samba-client	x86_64	4.22.4-12.el9_7	appstream	760 k

Upgrading:

libipa_hbac	x86_64	2.9.7-4.el9_7.1	baseos	33 k
libldb	x86_64	4.22.4-12.el9_7	baseos	180 k
libsmbclient	x86_64	4.22.4-12.el9_7	baseos	73 k
libsss_certmap	x86_64	2.9.7-4.el9_7.1	baseos	88 k
libsss_idmap	x86_64	2.9.7-4.el9_7.1	baseos	39 k
libsss_nss_idmap	x86_64	2.9.7-4.el9_7.1	baseos	43 k
libsss_sudo	x86_64	2.9.7-4.el9_7.1	baseos	33 k
libtalloc	x86_64	2.4.3-1.el9	baseos	33 k
libtdb	x86_64	1.4.13-1.el9	baseos	53 k
libtevent	x86_64	0.16.2-1.el9	baseos	50 k
libwbclient	x86_64	4.22.4-12.el9_7	baseos	42 k
samba-client-libs	x86_64	4.22.4-12.el9_7	baseos	5.3 M
samba-common	noarch	4.22.4-12.el9_7	baseos	173 k
samba-common-libs	x86_64	4.22.4-12.el9_7	baseos	104 k
sssd	x86_64	2.9.7-4.el9_7.1	baseos	25 k
sssd-ad	x86_64	2.9.7-4.el9_7.1	baseos	217 k
sssd-client	x86_64	2.9.7-4.el9_7.1	baseos	158 k
sssd-common	x86_64	2.9.7-4.el9_7.1	baseos	1.6 M
sssd-common-pac	x86_64	2.9.7-4.el9_7.1	baseos	94 k
sssd-ipa	x86_64	2.9.7-4.el9_7.1	baseos	286 k
sssd-kcm	x86_64	2.9.7-4.el9_7.1	baseos	107 k
sssd-krb5	x86_64	2.9.7-4.el9_7.1	baseos	70 k

Рис. 7: Этап 7



The image shows a terminal window with a dark background. At the top, there is a title bar with a window icon, the text "root@client:~", and search, menu, and close buttons. Below the title bar, there are two tabs: "root@client:~" (active) and "vagrant@client:~". The main area of the terminal displays XML configuration for a firewall service. The XML is as follows:

```
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>Samba Client</short>
  <description>This option allows you to access Windows file and printer sharing networks.
You need the samba-client package installed for this option to be useful.</description>
  <include service="netbios-ns"/>
  <port protocol="udp" port="138"/>
</service>
```

The last line of the XML, `/usr/lib/firewalld/services/samba-client.xml (END)`, is highlighted with a light blue background.

Рис. 8: Этап 8

```
1 # See smb.conf.example for a more detailed config file or
2 # read the smb.conf manpage.
3 # Run 'testparm' to verify the config is correct after
4 # you modified it.
5 #
6 # Note:
7 # SMB1 is disabled by default. This means clients without support for SMB2 or
8 # SMB3 are no longer able to connect to smbd (by default).
9
10 [global]
11     workgroup = USER-NET
12     security = user
13
14     passdb backend = tdbsam
15
16     printing = cups
17     printcap name = cups
18     load printers = yes
19     cups options = raw
20
21 [homes]
22     comment = Home Directories
23     valid users = %S, %D%w%S
24     browseable = No
25     read only = No
26     inherit acls = Yes
27
28 [printers]
29     comment = All Printers
30     path = /var/tmp
31     printable = Yes
32     create mask = 0600
33     browseable = No
34
35 [print$]
36     comment = Printer Drivers
37     path = /var/lib/samba/drivers
38     # printadmin is a local group
```

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Рис. 9: Этап 9

```
1 username=frank
2 password=houston237
```

Рис. 10: Этап 10

```

10 # units generated from this file.
11 #
12 UUID=c84cce45-9089-48d9-9617-2f1d1bd45fdd /                xfs      defaults    0 0
13 /swapfile none swap defaults 0 0
14 #VAGRANT-BEGIN
15 # The contents below are automatically generated by Vagrant. Do not modify.
16 vagrant /vagrant vboxsf uid=1000,gid=1000,_netdev 0 0
17 #VAGRANT-END
18
19 /server/sambashare /mnt/samba cifs
20 user,rw,uid=user_name,gid=sambagroup,
21 credentials=/etc/samba/smbusers,_netdev 0 0

```

Рис. 11: Этап 11

```

Added user user.
[root@server sambashare]# cd /vagrant/provision/server
[root@server server]# mkdir -p /vagrant/provision/server/smb/etc/samba
[root@server server]# cp -R /etc/samba/smb.conf /vagrant/provision/server/smb/etc/samba/
[root@server server]# cd /vagrant/provision/server
[root@server server]# touch smb.sh
[root@server server]# chmod +x smb.sh
[root@server server]# gedit smb
smb/    smb.sh
[root@server server]# gedit smb.sh

(gedit:9421): dconf-WARNING **: 16:21:35.722: failed to commit changes to dconf: Failed
to execute child process "dbus-launch" (No such file or directory)

```

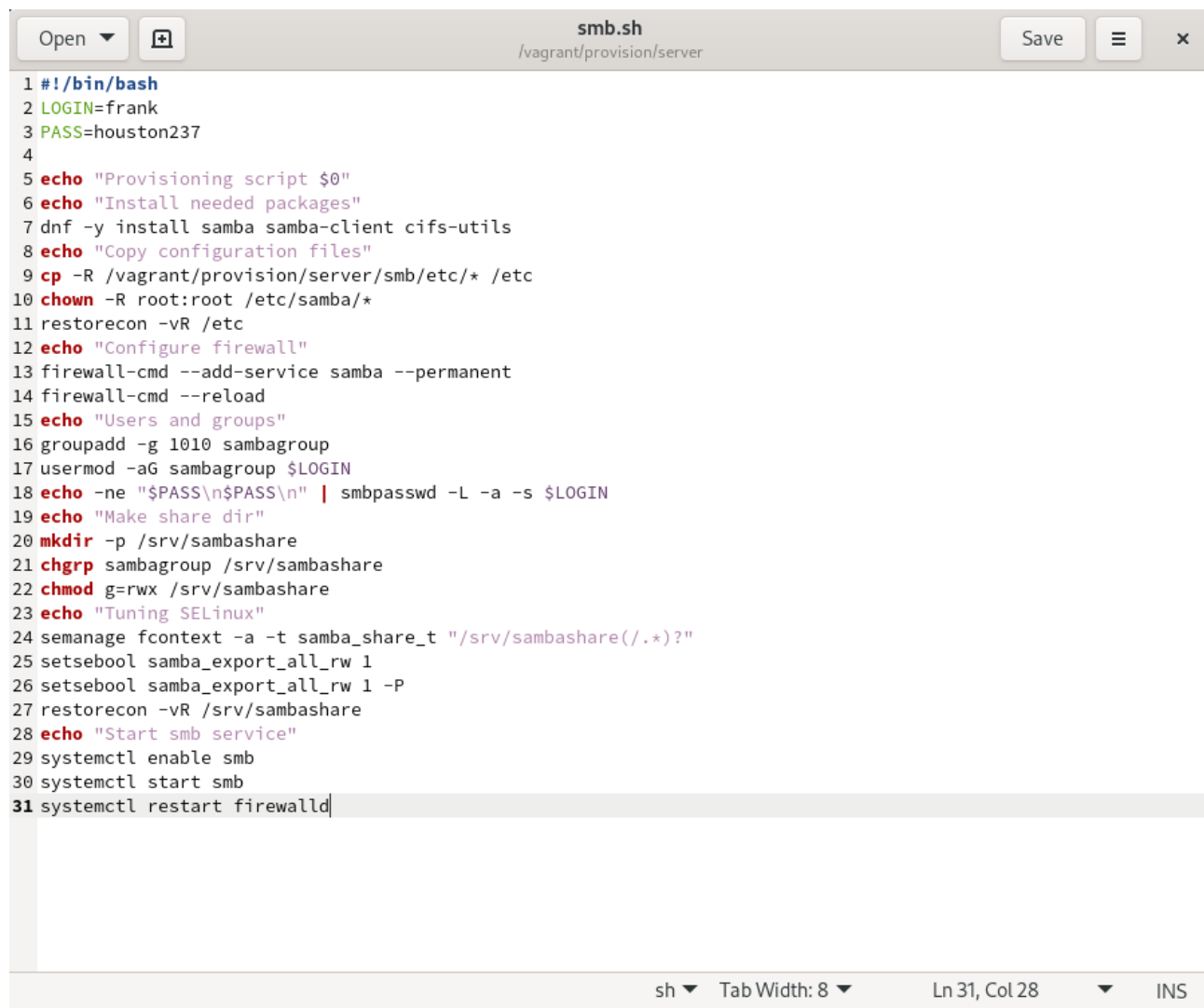
Рис. 12: Этап 12


```
1 #!/bin/bash
2 LOGIN=frank
3 PASS=houston237
4
5 echo "Provisioning script $0"
6 echo "Install needed packages"
7 dnf -y install samba samba-client cifs-utils
8 echo "Copy configuration files"
9 cp -R /vagrant/provision/server/smb/etc/* /etc
10 chown -R root:root /etc/samba/*
11 restorecon -vR /etc
12 echo "Configure firewall"
13 firewall-cmd --add-service samba --permanent
14 firewall-cmd --reload
15 echo "Users and groups"
16 groupadd -g 1010 sambagroup
17 usermod -aG sambagroup $LOGIN
18 echo -ne "$PASS\n$PASS\n" | smbpasswd -L -a -s $LOGIN
19 echo "Make share dir"
20 mkdir -p /srv/sambashare
21 chgrp sambagroup /srv/sambashare
22 chmod g=rwx /srv/sambashare
23 echo "Tuning SELinux"
24 semanage fcontext -a -t samba_share_t "/srv/sambashare(/.*)?"
25 setsebool samba_export_all_rw 1
26 setsebool samba_export_all_rw 1 -P
27 restorecon -vR /srv/sambashare
28 echo "Start smb service"
29 systemctl enable smb
30 systemctl start smb
31 systemctl restart firewalld
```

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Рис. 13: Этап 13

Этап выполнения работы №14



```
1 #!/bin/bash
2 LOGIN=frank
3 PASS=houston237
4
5 echo "Provisioning script $0"
6 echo "Install needed packages"
7 dnf -y install samba samba-client cifs-utils
8 echo "Copy configuration files"
9 cp -R /vagrant/provision/server/smb/etc/* /etc
10 chown -R root:root /etc/samba/*
11 restorecon -vR /etc
12 echo "Configure firewall"
13 firewall-cmd --add-service samba --permanent
14 firewall-cmd --reload
15 echo "Users and groups"
16 groupadd -g 1010 sambagroup
17 usermod -aG sambagroup $LOGIN
18 echo -ne "$PASS\n$PASS\n" | smbpasswd -L -a -s $LOGIN
19 echo "Make share dir"
20 mkdir -p /srv/sambashare
21 chgrp sambagroup /srv/sambashare
22 chmod g=rwx /srv/sambashare
23 echo "Tuning SELinux"
24 semanage fcontext -a -t samba_share_t "/srv/sambashare(/.*)?"
25 setsebool samba_export_all_rw 1
26 setsebool samba_export_all_rw 1 -P
27 restorecon -vR /srv/sambashare
28 echo "Start smb service"
29 systemctl enable smb
30 systemctl start smb
31 systemctl restart firewalld
```

Рис. 14: Этап 14

Этап выполнения работы №15

Этап выполнения работы №16

Этап выполнения работы №17

Выводы

В ходе выполнения лабораторной работы были приобретены практические навыки

Спасибо за внимание!

Вопросы?

```
(gedit:4874): dconf-WARNING **: 16:21:57.020: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
[root@client samba]# mount -a
mount: /etc/fstab: parse error at line 20 -- ignored
mount.cifs: bad UNC (/server/smbashare)
mount: (hint) your fstab has been modified, but systemd still uses
the old version; use 'systemctl daemon-reload' to reload.
mount: 0: mount point does not exist.
[root@client samba]# cd /vagrant/provision/client
[root@client client]# mkdir -p /vagrant/provision/client/smb/etc/samba
[root@client client]# cp -R /etc/samba/smb.conf /vagrant/provision/client/smb/etc/samba/
[root@client client]# cp -R /etc/samba/smbusers /vagrant/provision/client/smb/etc/samba/
[root@client client]# cd /vagrant/provision/client
[root@client client]# touch smb.sh
[root@client client]# chmod +x smb.sh
[root@client client]# gedit smb.sh

(gedit:4959): dconf-WARNING **: 16:32:55.868: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:4959): dconf-WARNING **: 16:32:55.874: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:4959): dconf-WARNING **: 16:32:56.084: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:4959): dconf-WARNING **: 16:32:56.110: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:4959): dconf-WARNING **: 16:32:56.110: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
```

Рис. 15: Этап 15

```
*smb.sh
/vagrant/provision/client

Could not save the file "/vagrant/provision/client/smb.sh".
Unexpected error: Error renaming temporary file: Text file busy

1 #!/bin/bash
2 LOGIN=user
3 echo "Provisioning script $0"
4
5 mkdir -p /mnt/samba
6 echo "Install needed packages"
7 dnf -y install samba-client cifs-utils
8 echo "Copy configuration files"
9 cp -R /vagrant/provision/client/smb/etc/* /etc
10 chown -R root:root /etc/samba/*
11 restorecon -vR /etc
12 echo "Configure firewall"
13 firewall-cmd --add-service samba-client --permanent
14 firewall-cmd --reload
15 echo "Users and groups"
16 groupadd -g 1010 sambagroup
17 usermod -aG sambagroup $LOGIN
18 echo "Mounting dirs"
19 mkdir -p /srv/smbashare
20 echo "//server/smbashare /mnt/samba cifs
21 user,rw,credentials=/etc/samba/smbusers,uid=user,
22 gid=sambagroup,_netdev 0 0" >> /etc/fstab
23 ↵
24 ↵
25 restorecon -vR /etc
26 umount /mnt/samba
27 mount /mnt/samba
```

Рис. 16: Этап 16

```

1 #!/bin/bash
2 LOGIN=user
3 echo "Provisioning script $0"
4
5 mkdir -p /mnt/samba
6 echo "Install needed packages"
7 dnf -y install samba-client cifs-utils
8 echo "Copy configuration files"
9 cp -R /vagrant/provision/client/smb/etc/* /etc
10 chown -R root:root /etc/samba/*
11 restorecon -vR /etc
12 echo "Configure firewall"
13 firewall-cmd --add-service samba-client --permanent
14 firewall-cmd --reload
15 echo "Users and groups"
16 groupadd -g 1010 sambagroup
17 usermod -aG sambagroup $LOGIN
18 echo "Mounting dirs"
19 mkdir -p /srv/sambashare
20 echo "///server/sambashare /mnt/samba cifs
21 user,rw,credentials=/etc/samba/smbusers,uid=user,
22 gid=sambagroup,_netdev 0 0" >> /etc/fstab
23 ↵
24 ↵
25 restorecon -vR /etc
26 umount /mnt/samba
27 mount /mnt/samba

```

Рис. 17: Этап 17